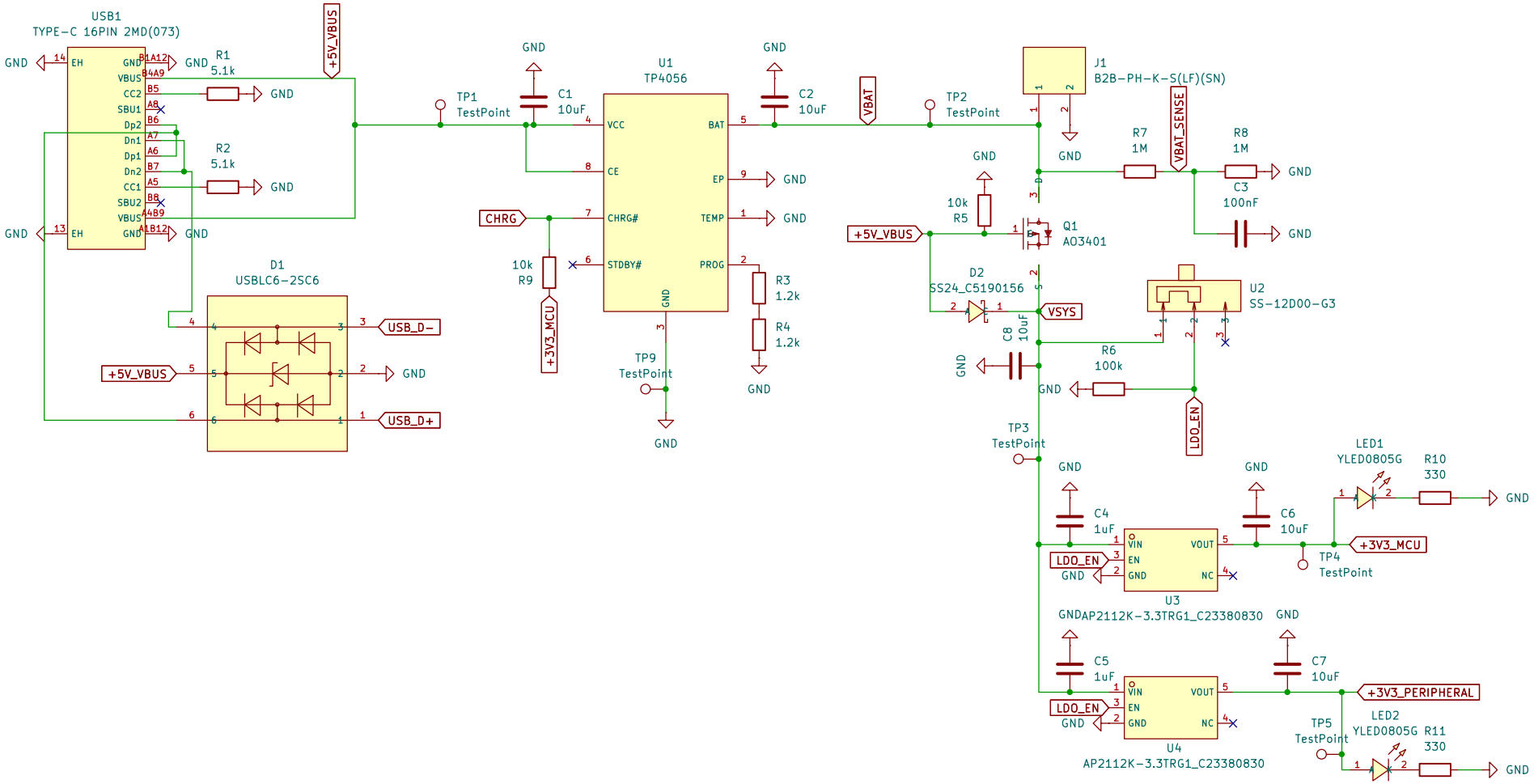
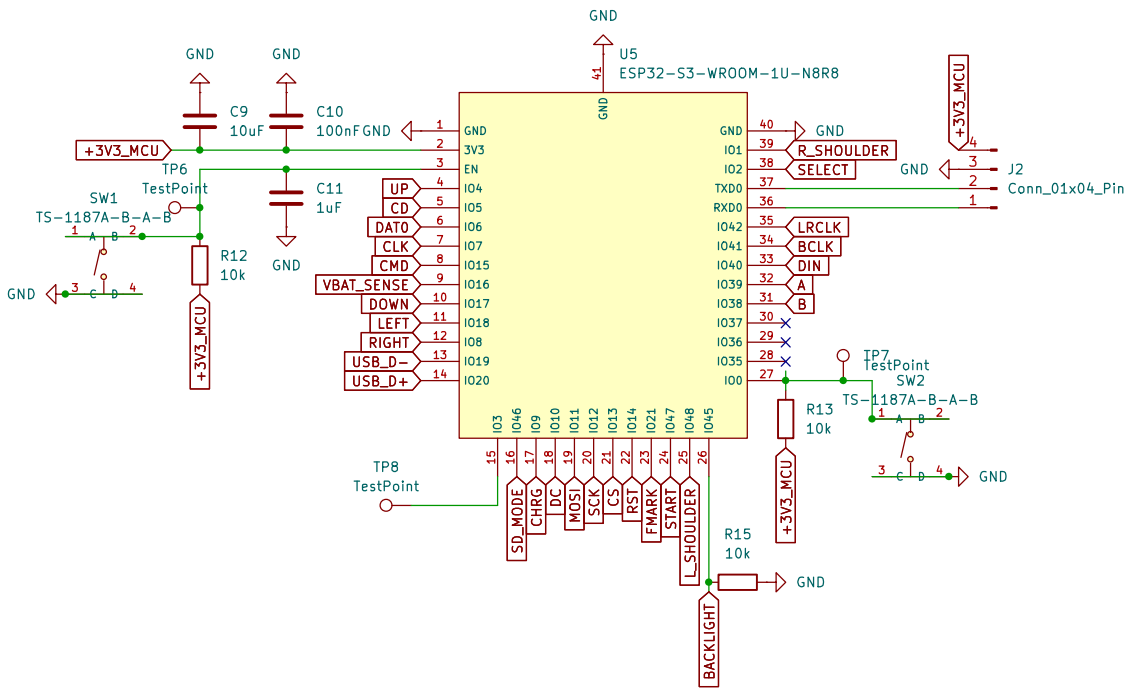
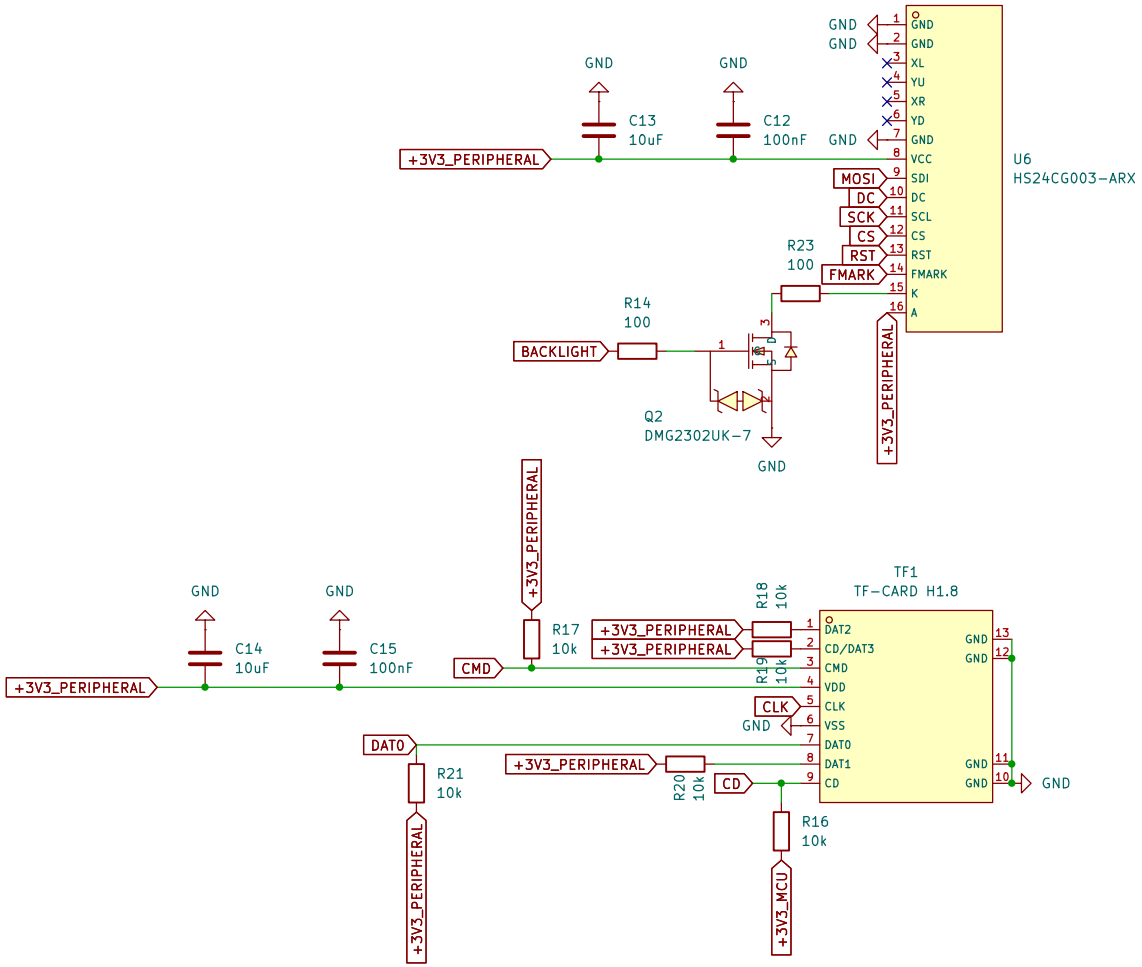
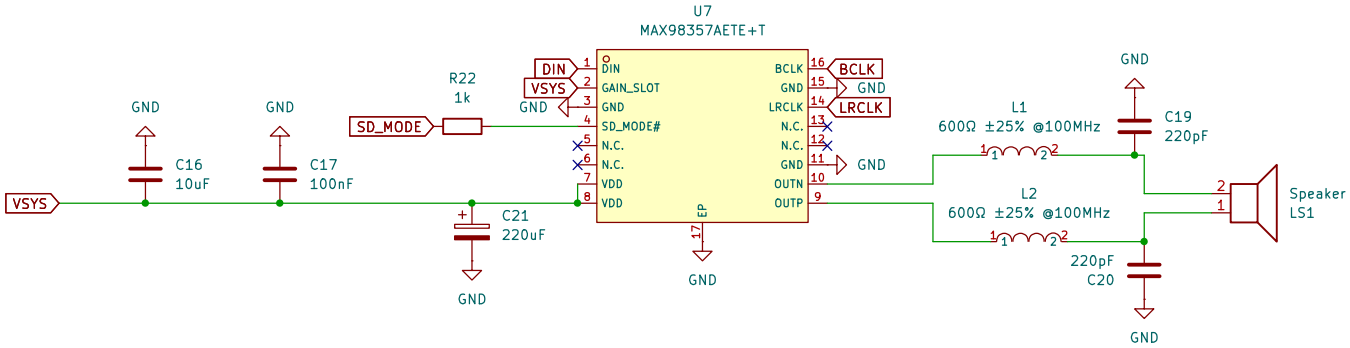
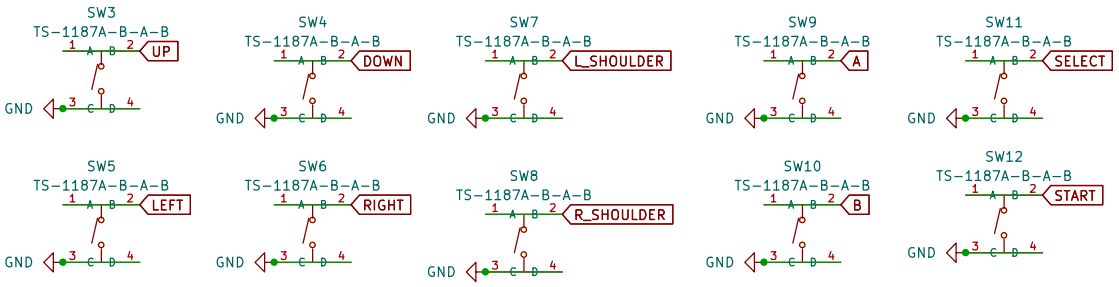

USB1
TYPE-C 16PIN 2MD(073)
R1
5.1k
R2
5.1k
+5V_VBUS
D1
USBLC6-2SC6
USB_D-
USB_D+
TP1
TestPoint
C1
10uF
U1
TP4056
C2
10uF
VBAT
CHRG
10k
R9
+3V3_MCU
R3
1.2k
R4
1.2k
TP9
TestPoint
TP2
TestPoint
J1
B2B-PH-K-S(LF)(SN)
R7
1M
R8
1M
VBAT_SENSE
C3
100nF
10k
R5
Q1
AO3401
D2
SS24_C5190156
VSYS
C8
10uF
U2
SS-12D00-G3
R6
100k
LDO_EN
TP3
TestPoint
C4
1uF
C6
10uF
LED1
YLED0805G
R10
330
TP4
TestPoint
U3
AP2112K-3.3TRG1_C23380830
C5
1uF
C7
10uF
+3V3_PERIPHERAL
TP5
TestPoint
LED2
YLED0805G
R11
330
U4
AP2112K-3.3TRG1_C23380830
GND


GND
U5
ESP32-S3-WROOM-1U-N8R8
GND
GND
C9
10uF
C10
100nF
GND
+3V3_MCU
TP6
TestPoint
SW1
TS-1187A-B-A-B
C11
1uF
R12
10k
GND
+3V3_MCU
UP
CD
DAT0
CLK
CMD
VBAT_SENSE
DOWN
LEFT
RIGHT
USB_D-
USB_D+
GND
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19
IO20
GND
IO1
IO2
TXD0
RXD0
IO42
IO41
IO40
IO39
IO38
IO37
IO36
IO35
IO0
R_SHOULDER
SELECT
LRCLK
BCLK
DIN
A
B
+3V3_MCU
GND
J2
Conn_01x04_Pin
TP7
TestPoint
SW2
TS-1187A-B-A-B
R13
10k
+3V3_MCU
GND
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
TP8
TestPoint
SD_MODE
CHRG
DC
MOSI
SCK
CS
RST
FMARK
START
L_SHOULDER
R15
10k
GND
BACKLIGHT


GND
GND
C13
10uF
GND
GND
C12
100nF
+3V3_PERIPHERAL
U6
HS24CG003-ARX
GND
GND
XL
YU
XR
YD
GND
VCC
MOSI
DC
SCK
CS
RST
FMARK
K
A
SDI
DC
SCL
CS
RST
FMARK
R23
100
R14
100
BACKLIGHT
Q2
DMG2302UK-7
GND
+3V3_PERIPHERAL
GND
GND
C14
10uF
C15
100nF
+3V3_PERIPHERAL
+3V3_PERIPHERAL
R17
10k
CMD
R18
10k
R19
10k
+3V3_PERIPHERAL
+3V3_PERIPHERAL
TF1
TF-CARD H1.8
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
CD
GND
GND
GND
GND
GND
CLK
GND
DAT0
R21
10k
+3V3_PERIPHERAL
+3V3_PERIPHERAL
R20
10k
CD
R16
10k
+3V3_MCU
GND


U7
MAX98357AETE+T
DIN
GAIN_SLOT
GND
SD_MODE#
N.C.
N.C.
VDD
VDD
EP
BCLK
GND
LRCLK
N.C.
N.C.
GND
OUTN
OUTP
17
R22
1k
SD_MODE
DIN
VSYS
GND
BCLK
GND
LRCLK
GND
VSYS
C16
10uF
C17
100nF
C21
220uF
L1
600Ω ±25% @100MHz
L2
600Ω ±25% @100MHz
C19
220pF
220pF
C20
Speaker
LS1


SW3
TS-1187A-B-A-B
UP
GND
SW4
TS-1187A-B-A-B
DOWN
GND
SW7
TS-1187A-B-A-B
L_SHOULDER
GND
SW9
TS-1187A-B-A-B
A
GND
SW11
TS-1187A-B-A-B
SELECT
GND
SW5
TS-1187A-B-A-B
LEFT
GND
SW6
TS-1187A-B-A-B
RIGHT
GND
SW8
TS-1187A-B-A-B
R_SHOULDER
GND
SW10
TS-1187A-B-A-B
B
GND
SW12
TS-1187A-B-A-B
START
GND